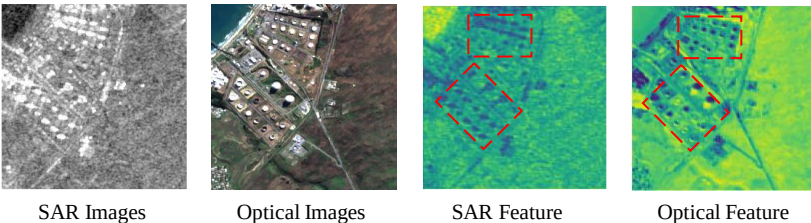


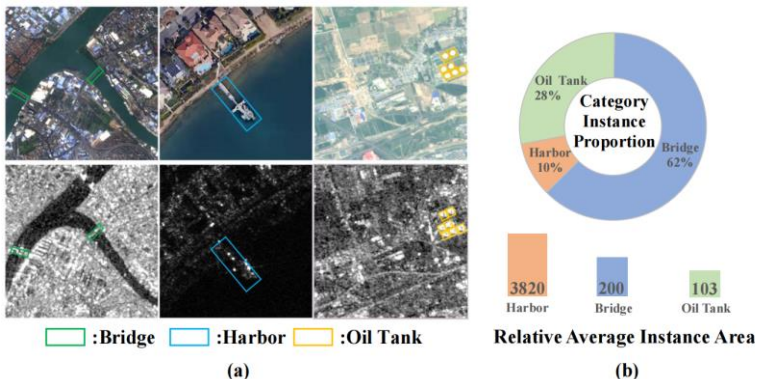
Reporter: Chao Wang. Member: Lei Luo*, Wenxuan Fang, Jian Yang

Motivation



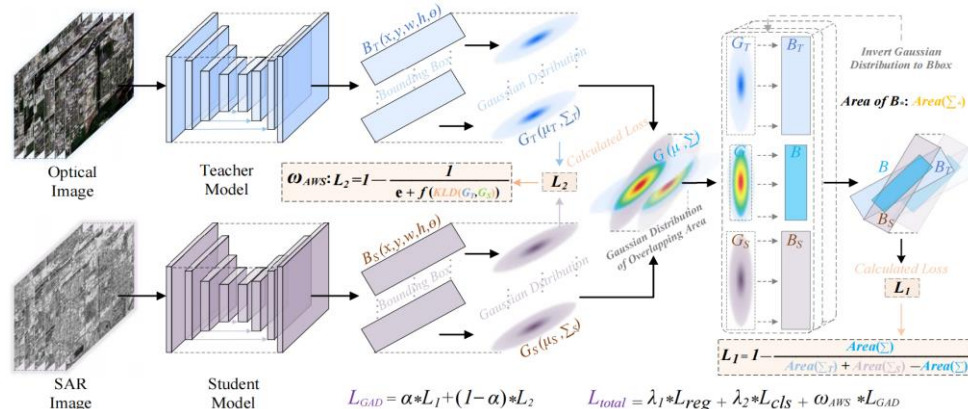
- **Imaging Mechanism:** SAR images represent the magnitude of **reflected energy**, leading to poor object boundary continuity.
- **Orientation Detection:** Accurately detecting **object angles** in SAR images is difficult due to the imaging mechanism.
- Existing datasets like OGSOD-1.0 lack high-quality oriented labels, hindering progress in this field.

OGSOD-2.0 Dataset



- Due to the lack of **high-quality** oriented labels and instances in the dataset **OGSOD-1.0**.
- Therefore, we expand **OGSOD-1.0** dataset to create an oriented object detection dataset (**OGSOD-2.0**) by adding high-quality oriented labels and images.
- The **OGSOD-2.0** dataset contains **20,359** pairs of optical and SAR images, and more than **54,000 high-quality oriented labels**. Fig. (a) shows an examples of an image pairs, and Fig. (b) shows the statistics of the instance.

Methodology-1



Methodology-2

$$Area(\Sigma_s) = 2^2 \sqrt{\prod eig(\Sigma_s)} = 2^2 \cdot \sqrt{|\Sigma_s|}$$

$$L_1 = 1 - \frac{Area(\Sigma)}{Area(\Sigma_t) + Area(\Sigma_s) - Area(\Sigma)}$$

$$L_2 = 1 - \frac{1}{e + f(KLD(G_t, G_s))}, e \geq 1$$

$$L_{GAD} = \alpha * L_1 + (1 - \alpha) * L_2$$

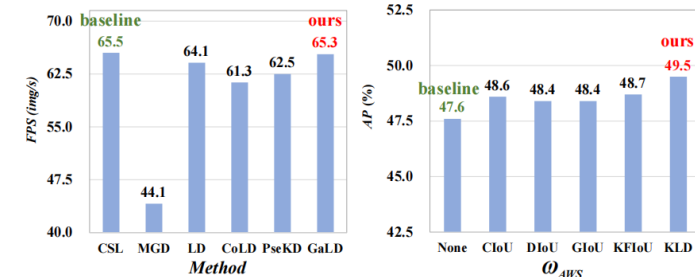
- We convert the **oriented bounding box** into a **Gaussian distribution** with angle information. Then, we use the **KLD** to reduce the **distance** between the Gaussian distributions generated by the teacher and student networks.

Experimental Evaluation

Method	Venue	Backbone	Dataset	KD Types	OGSOD-2.0 (SAR)				
					Oil Tank ↑	Bridge ↑	Harbor ↑	AP ₅₀ ↑	AP ↑
Rotate Focus [16]	ICCV'2019	ResNet-50	SAR	-	51.7	69.6	90.8	70.7	44.9
Gliding Vertex [17]	TPAMI'2020	ResNet-50	SAR	-	44.9	78.2	89.1	70.8	40.4
CSL (Baseline) [18]	ECCV'2020	CSPDarkNet-M	SAR	-	39.8	84.6	97.0	73.8	40.0
Oriented R-CNN [19]	ICCV'2021	ResNet-50	SAR	-	35.4	70.2	90.2	65.3	37.2
S2A-Net [20]	TGRS'2021	ResNet-50	SAR	-	68.9	79.9	90.9	79.9	41.8
GWD [21]	ICML'2021	ResNet-50	SAR	-	54.8	77.8	90.4	74.4	44.4
KFIoU [22]	ICLR'2023	ResNet-50	SAR	-	40.2	79.2	90.6	70.0	35.5
PSC [23]	CVPR'2023	CSPDarkNet-M	SAR	-	37.3	73.8	93.6	68.2	34.5
Mimic [24]	TPAMI'2021	CSPDarkNet-M	Optical-SAR	F	52.7	87.0	96.1	78.6	43.1
CWD [25]	CVPR'2021	ResNet-50	Optical-SAR	F	53.7	87.5	96.2	79.1	43.1
MGD [26]	ECCV'2022	ResNet-50	Optical-SAR	F	50.7	86.1	94.9	77.2	42.2
LD [27]	CVPR'2022	CSPDarkNet-M	Optical-SAR	L	46.8	93.2	98.4	79.5	44.8
CoLD [15]	TGRS'2023	CSPDarkNet-M	Optical-SAR	L	50.3	93.3	99.1	80.9	46.0
PseKD [28]	ICASSP'2024	CSPDarkNet-M	Optical-SAR	L	58.5	90.1	97.5	82.1	46.4
GaLD (Ours)	-	CSPDarkNet-M	Optical-SAR	L	65.8	91.5	98.2	85.2	49.5

¹ Best and ² Second best. Higher AP₅₀ and AP are better.

Comparison of FPS and AP of different methods



- Compared to the baseline, our proposed GaLD achieves the highest **AP** and **FPS** with introducing **KLD**. This result proves that there is **low-quality knowledge** in the **teacher network** and that we need to use a strategy to focus on high-quality knowledge.

Ablation Experiment

TABLE II
ABLATION EXPERIMENTS ON THE OGSOD-2.0 DATASET.

GAD_{L_1}	GAD_{L_2}	ω_{AWS}	AP ₅₀ ↑	AP ↑
-	-	-	73.8	40.0
✓	-	-	81.4	47.0
-	✓	-	81.5	45.2
✓	✓	-	84.7	47.6
✓	✓	✓	85.2	49.5

- The distillation strategy of L_1 and L_2 increases the **AP** by **7.0%** and **5.2%**, respectively. $L_1 + L_2$ and $L_1 + L_2 + \omega_{AWS}$ increases the **AP** by **7.6%** and **9.5%**, respectively.

Detection Visual Results

